EXPERIMENT-7a

S.Vishal 240701600

Develop Low-Fidelity Paper Prototypes for a Banking App and Convert Them into Digital Wireframes Using Pencil Project

AIM

The aim of this experiment is to develop low-fidelity paper prototypes for a banking application and convert them into digital wireframes using Pencil Project, an open-source wireframing tool. The experiment covers the complete UI prototyping workflow from initial sketching to a structured digital wireframe with screen navigation.

Tools Used

- Pencil Project – for creating digital wireframes and screen navigation flows
- Paper and Pencil – for initial low-fidelity sketches
- Tool Link: https://pencil.evolus.vn/

Procedure

Step 1: Create Low-Fidelity Paper Prototypes

1. Define the Purpose and Features:

The banking app is designed to allow users to securely access their account, perform financial transactions, and manage their banking activities. The following core features were identified:

- Login / Authentication Screen – Username and password entry with secure access.
- Dashboard – Overview of account balance, recent transactions, and quick action buttons.
- Fund Transfer – Enter beneficiary details and transfer money securely.
- Bill Payments – Browse payment categories and pay utility bills directly.
- Transaction History – View a filterable list of all past transactions.
- Profile – View and manage personal account details and settings.

2. Sketch Basic Layouts:

Plain paper and pencils were used to sketch rough layouts for each screen. The focus was on placing primary UI elements without worrying about visual styling. Key elements sketched include:

- Login Screen: Logo placeholder, two input fields, login button, and register link.
- Dashboard: Balance card, quick-action icons (Transfer, Pay Bills, Deposit, History), and a recent transactions list.
- Fund Transfer: Sender account selector, beneficiary account field, amount input, IFSC code, and confirm button.
- Bill Payments: Category icon grid (Electricity, Mobile, Internet, Water, DTH, Rent) and a recent billers section.
- Transaction History: Filter tabs (All / Credit / Debit / Transfers) and a scrollable list of transactions with amounts.

- Profile: User avatar area, account details card, and settings menu items.

3. Iterate and Refine:

The initial paper sketches were reviewed with peers and stakeholders. Feedback was collected and the layouts were refined before moving to digital wireframes:

- Feedback: The dashboard needed clearer visual separation between the balance card and the quick-action area.
- Improvement: Added a section label "Quick Actions" and increased spacing between the balance card and the action icons.
- Feedback: The Fund Transfer form was cluttered with all fields stacked vertically.
- Improvement: Amount and IFSC fields were placed side by side to reduce scrolling.

Step 2: Convert Paper Prototypes to Digital Wireframes Using Pencil Project

1. Install Pencil Project:

Pencil Project was downloaded from the official website (https://pencil.evolus.vn/) and installed on the system. The tool was launched and a new blank project was created.

2. Create a New Document:

A new document was created in Pencil Project titled "Banking App Wireframes". The document was configured with a mobile canvas size (360 × 640 px) to match standard smartphone dimensions.

3. Add Screens:

Six pages were added to the document, one for each screen of the banking application:

- Page 1 – Login Screen
- Page 2 – Dashboard
- Page 3 – Fund Transfer
- Page 4 – Bill Payments
- Page 5 – Transaction History
- Page 6 – Profile

4. Use Stencils and Shapes:

Pencil Project's built-in stencil library was used to place UI elements on each canvas. The following stencil categories were used:

- Android / iOS Mockups – for status bar and navigation bar
- Form Elements – for input fields, buttons, checkboxes, and dropdowns
- Common Shapes – for cards, dividers, and icon placeholders
- Typography stencils – for labels, headings, and placeholder text

5. Organise and Align:

Elements were arranged and aligned on each screen to match the refined paper sketches. Guidelines and snap-to-grid features in Pencil Project were used to ensure consistent spacing and alignment across all screens.

6. Link Screens:

Connectors and hyperlinks were used in Pencil Project to link UI elements across screens, creating a navigational flow:

- Login button → Dashboard

- Register link → Register Screen (noted as future scope)
- Transfer icon on Dashboard → Fund Transfer Screen
- Pay Bills icon on Dashboard → Bill Payments Screen
- History icon on Dashboard → Transaction History Screen
- Profile icon in navigation bar → Profile Screen
- Back button on all sub-screens → Dashboard

7. Add Annotations:

Annotations were added to key elements across all screens to explain their functionality:

- Login Screen: "Password field uses masking; Forgot Password triggers OTP flow"
- Dashboard: "Balance card shows real-time account balance fetched from backend API"
- Fund Transfer: "IFSC Code field auto-validates on entry; Confirm button triggers 2FA"
- Bill Payments: "Categories fetch registered billers; Pay Now button opens payment gateway"
- Transaction History: "Tabs filter transactions dynamically; entries are paginated"
- Profile: "Logout clears session token and redirects to Login screen"

8. Export Wireframes:

After completing all six screens, the wireframes were exported from Pencil Project in PNG format for documentation and sharing. The final exported screens are shown below.

Digital Wireframes – Output Screens

The following screenshots show the six digital wireframes created in Pencil Project for the Banking Application:

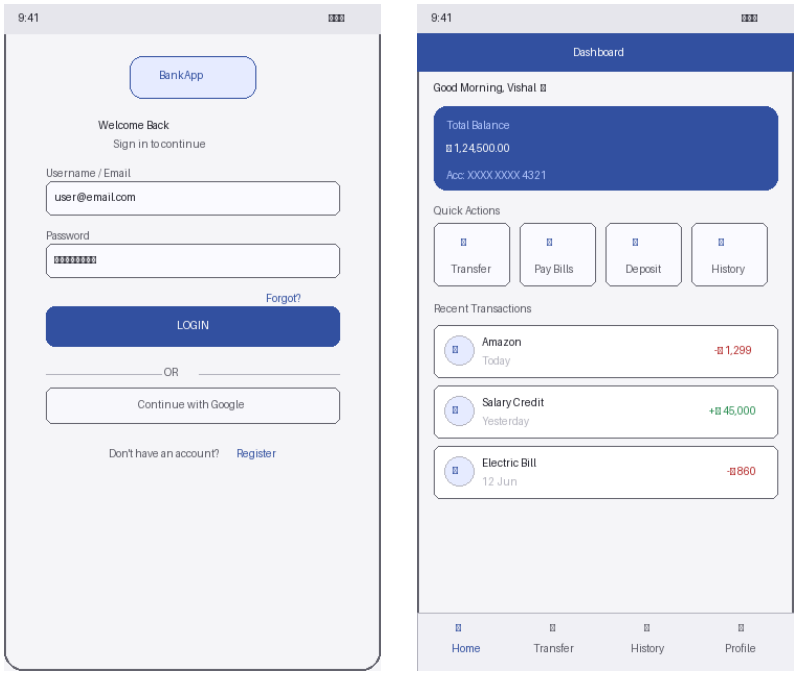

Fig 1: Login Screen (left) | Dashboard Screen (right)

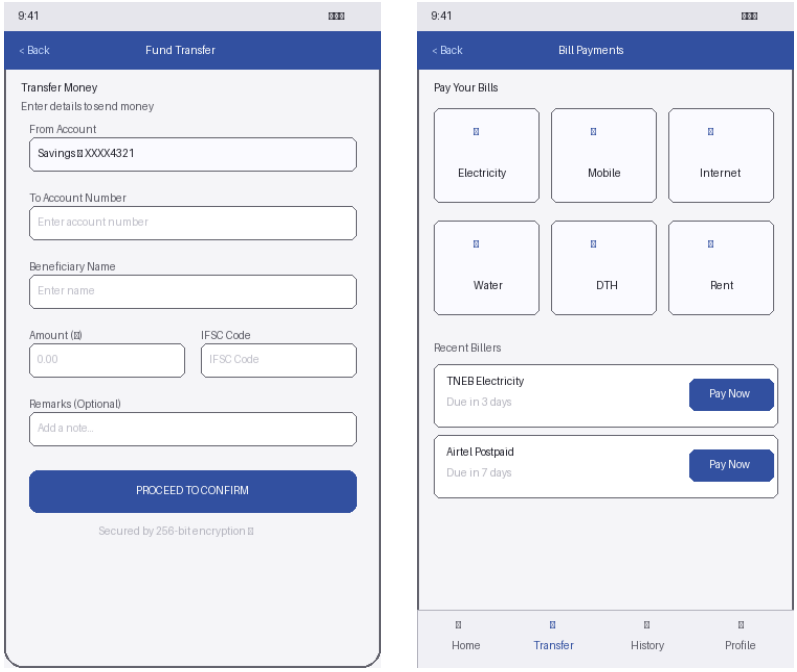

Fig 2: Fund Transfer Screen (left) | Bill Payments Screen (right)

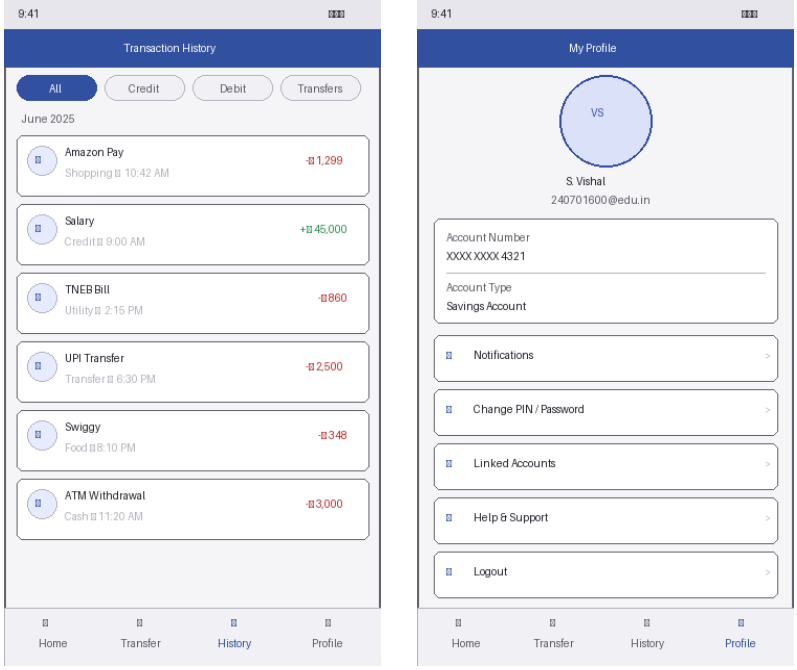

Fig 3: Transaction History Screen (left) | Profile Screen (right)

Screen Descriptions

Screen 1: Login Screen

The Login screen is the entry point of the banking application. It contains a logo area at the top for branding, followed by input fields for Username/Email and Password. A "Forgot Password" link is placed

beside the password field for account recovery. The primary LOGIN button is prominently placed. A divider with "OR" separates the option to continue with Google OAuth. A link at the bottom allows new users to navigate to the Registration screen.

Screen 2: Dashboard Screen

The Dashboard is the main home screen after successful login. It displays a personalised greeting at the top. A highlighted balance card shows the total account balance and masked account number. Below it, a Quick Actions row provides icon shortcuts for Fund Transfer, Bill Payments, Deposit, and Transaction History. The lower section displays a Recent Transactions list showing the latest entries with merchant name, category, date, and colour-coded debit/credit amounts. A bottom navigation bar provides persistent access to all major sections.

Screen 3: Fund Transfer Screen

The Fund Transfer screen allows users to send money to another account. It contains a "From Account" field pre-filled with the user's active account. The beneficiary details section includes fields for Account Number, Beneficiary Name, Amount, IFSC Code, and an optional Remarks field. The "Proceed to Confirm" button initiates the transfer flow. A security note at the bottom reassures users about encryption.

Screen 4: Bill Payments Screen

The Bill Payments screen is organised into two sections. The top section shows a grid of payment category icons (Electricity, Mobile, Internet, Water, DTH, Rent) for easy category selection. The lower section lists Recent Billers — saved billers from the user's history — showing the biller name, due amount, and a quick "Pay Now" button for one-tap payment.

Screen 5: Transaction History Screen

The Transaction History screen provides a complete log of the user's financial activity. Filter tabs at the top allow users to view All transactions or filter by Credit, Debit, or Transfers. Each transaction entry displays the merchant or payee name, category, date and time, and the transaction amount in green (credit) or red (debit) colour coding for quick readability.

Screen 6: Profile Screen

The Profile screen displays the user's personal information and account settings. An avatar placeholder at the top shows the user's initials. Below it, an Account Information card shows the masked account number and account type. A menu list provides access to Notifications, Change PIN/Password, Linked Accounts, Help & Support, and Logout. Each menu item has a right-arrow indicator for navigation.

Result

The experiment was completed successfully. Low-fidelity paper prototypes were created for six screens of a Banking Application: Login, Dashboard, Fund Transfer, Bill Payments, Transaction History, and Profile. These paper sketches were then converted into structured digital wireframes using Pencil Project. All screens were designed with appropriate UI stencils, annotated for functionality, linked together with navigation flows, and exported as PNG images for documentation.

Conclusion

This experiment demonstrated the complete UI prototyping workflow from low-fidelity paper sketching to structured digital wireframes. Paper prototyping helped in quickly exploring layout ideas and gathering early feedback without investing time in detailed design. Pencil Project proved to be an effective tool for converting those rough ideas into organised, annotated wireframes with screen navigation. Together, these techniques support efficient and user-centred interface design by identifying usability issues early in the development process, saving time and cost in later stages.